A Hundred Feeders, One Bottleneck: Where the Queue Forms on Avenida Lázaro Cárdenas, Monterrey

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October 12, 2026

Abstract

Avenida Lázaro Cárdenas links Monterrey with San Pedro Garza García and is congested every peak. A common explanation counts lanes: over a hundred feeder lanes pour into an avenue with five or six lanes per direction. We test that explanation as a queueing problem. From an OpenStreetMap extract (2026-10-12) we build an inventory of the 10.0 km corridor: 99 street approaches (plus 110 driveways and plaza entrances) deliver 133 feeder lanes (264 with driveways) into 5 corridor lanes per direction, a ratio of about 13:1. These figures fall inside the unsourced ranges that motivated the study. We model each direction as a tandem queue, validate a vehicle-level simulator against Erlang C, M/D/1 and Webster's signal delay (mean error 3.2%), and run a Cell Transmission Model whose total delay at a single bottleneck reproduces the fluid queue exactly. Feeder lanes deliver on average 151 veh/h per lane, or 8% of a lane's saturation flow, so lane counts overstate the pressure by an order of magnitude. Across a Monte Carlo over every demand and control assumption, the local lane ratio never points to the segment where the queue forms (hit rate 0%), and feeder lanes alone never do either. Steady-state utilisation ρ finds the bottleneck in 100% of congested draws and separates congested from free-flowing runs perfectly (AUC 1.00). Geometry alone (minimum capacity) does as well, because the bottleneck is a $5 \to 3$-lane merge at about 4.4 km, between the Alfonso Reyes overpass and the Arroyo Seco bridge, where the frontage road joins the express lanes. Filling that gap moves the queue elsewhere. A field estimate at Alfonso Reyes (570–770 veh/h joining towards San Pedro, against 2,100 assumed) lowers westbound utilisation at the merge from 1.04 to 0.88. Over a working day (7:00–20:00) with tidal demand, heading towards San Pedro in the morning, the westbound direction stays below capacity even at 7:00 ($\rho = 0.88$). That matches the observer's report of little or no morning queue towards San Pedro. The eastbound queue forms at 15:44 and is still there at 20:00, the end of the simulated day. At base parameters the eastbound queue reaches 6.0 km and costs 15,240 veh h per peak. How much of that is spillback into exits is the largest single uncertainty (from 4,170 to 61,510 veh h). An extra lane over 600 m at the merge changes delay by a median -37% and helped in every draw. Ramp metering, consolidating minor accesses and signal retiming give mixed or negligible results. The practical rule: count capacity, not lanes, and look for the merge.

1 Introduction

Avenida Lázaro Cárdenas runs about ten kilometres from the Garza Sada and Carretera Nacional interchange in south Monterrey to Avenida Gómez Morín in San Pedro Garza García. It carries traffic between the residential south and the Valle Oriente business district, and its peaks are long. The usual explanation goes like this: dozens of streets, ramps and plaza entrances feed the avenue, and if you add up their lanes you get well over a hundred, all emptying into five or six lanes per direction.

That argument is a lane count. A queueing system does not care how many lanes arrive. It cares how many vehicles per hour arrive and how many per hour can leave, at each point where they meet. A one-lane residential street that sends a hundred vehicles an hour is not the same as a lane of Garza Sada sending a thousand. A queue forms where demand exceeds capacity, and it backs up from there. So we ask:

> *Where does the queue form on Lázaro Cárdenas, and is that place predicted by the ratio of feeder lanes to corridor lanes, or by the utilisation (demand over capacity) of each merge?*

We expected utilisation to win, and we set a falsification rule before running anything: if the lane ratio finds the simulated bottleneck at least as often as utilisation does, the hypothesis fails.

The study makes four contributions. (i) An inventory of the corridor from OpenStreetMap: every approach that can deliver vehicles, its lanes, class and side, with internal express-to-frontage transfers removed (Section 3). Its totals turn out to agree with the figures that motivated the work. (ii) A queueing model at three levels: closed forms validated by a vehicle-level simulator, a steady-state tandem model, and a Cell Transmission Model

with merges, partially FIFO diverges and capacity drop (Section 4). (iii) A Monte Carlo over every assumption we could not measure, scoring five predictors of where the queue forms (Section 5). (iv) Four interventions evaluated where the queue actually forms (Section 6).

2 Related work

Queues and signals. Steady-state delay at a server follows from utilisation: exactly for M/M/1 and M/M/c (Erlang C) [12], through Kingman's heavy-traffic formula for G/G/1 [11], and through the Allen–Cunneen correction for many servers [1]. Webster [19] derived the delay at a fixed-time signal, and the Highway Capacity Manual [18] extends it to oversaturation over a finite period. When demand exceeds capacity only during a peak, the right tool is the deterministic fluid queue built from cumulative curves [14].

Kinematic waves. The LWR model [13, 17] treats traffic as a compressible fluid. Daganzo's Cell Transmission Model (CTM) [6, 7] discretises it with merge and diverge rules. Newell [15] showed that, for a triangular fundamental diagram, a spatial queue causes the same total delay as a point queue at the bottleneck, which gives us an exact check. Empirical work locates queues at active bottlenecks, the boundary between queued traffic upstream and freely flowing traffic downstream [4]. Lane drops are a classic case [2], and the queue's discharge rate falls by a few percent once the queue forms (capacity drop) [10, 5]. Local ramp metering with ALINEA [16] uses that drop: it keeps the bottleneck just below critical density.

Open street data. OpenStreetMap [9] is the standard open source of street geometry, and tools such as OSMnx [3] turn it into graphs. Lane tags are common on arterials but sparse on local streets, which our inventory reports explicitly.

Delta. We do not propose a new traffic model. We put a widely repeated lane-count argument about one specific Monterrey corridor to a falsifiable test, using open data and standard, validated models, and we separate the part of the answer that comes from geometry from the part that comes from assumed demand.

3 The corridor in OpenStreetMap

Extraction. We download the street network in 0.01-degree tiles along the corridor from the OpenStreetMap API (the Overpass mirrors were saturated on the day). The corridor consists of the `trunk` (express) and `secondary`/`tertiary` (frontage) ways named Lázaro Cárdenas that lie within 120 m of the centre line. We assign each one-way way to a direction by its travel direction along the chainage. Every other street way that touches a corridor node is a candidate feeder. One-way tags decide whether it can deliver vehicles (an approach), receive them (a departure), or both. Chains of unnamed links that only join the corridor to itself (express–frontage transfers, U-turns, median crossings) are internal; we found 52 such ways and removed them. A link that does reach the outside takes the demand class of the most important road it reaches, so a ramp that only leads to a parking aisle counts as a service access.

What feeds the avenue. Over 10.0 km we find 226 approaches. 110 of them are `service` ways (driveways, plaza and parking entrances), and 17 are retornos (U-turns and median crossings that bring vehicles from the opposite carriageway, 21 lanes; each is also an exit of the direction it leaves). The other 99 are streets: 20 major (trunk, primary, secondary and their links), 26 tertiary and 53 residential. Together the streets bring 133 feeder lanes (264 including driveways). 193 approaches end at the corridor (T-junctions, ramps); 33 cross it. The cross-section is 3 express lanes plus two or three frontage lanes, for a median of 5 lanes per direction (range 2–7).

The conversation that motivated this project estimated 80–100 accesses, 130–160 feeder lanes and 5–6 lanes per direction, without giving sources. The street counts land inside all three ranges. The lane ratio is real: about 13:1 without driveways and 26:1 with them. The question is whether it means anything.

Data quality. All corridor ways carry a `lanes` tag (100%). Only 28% of the feeder approaches do; the rest take a class default (one lane per direction for residential and service, two for tertiary). OpenStreetMap marks only 4 traffic signals on the corridor, all at Alfonso Reyes. That is surely incomplete, so we also treat as signalised every frontage junction fed by a major street or a tertiary street with at least two lanes. This gives 7 signalised cells westbound and 8 eastbound. The model's main bottleneck depends on one mapped feature, and we test it directly (Section 5): between 4.36 and 4.55 km the frontage road merges into the express lanes through a `secondary_link` and reappears only after the stretch, so the corridor narrows from five lanes to three between the Alfonso Reyes overpass and the bridge over the Canal del Arroyo Seco.

One field estimate. Alfonso Reyes reaches the corridor with four lanes from the south. Two of them join Lázaro Cárdenas towards San Pedro (westbound). While Alfonso Reyes has red, only one of them moves, through a continuous turn; on green both move. An observer on site estimated 7–10 vehicles per minute on red and 14–18 on green. Weighting by the model's green share for the cross street gives 570–770 veh/h for the whole junction (named

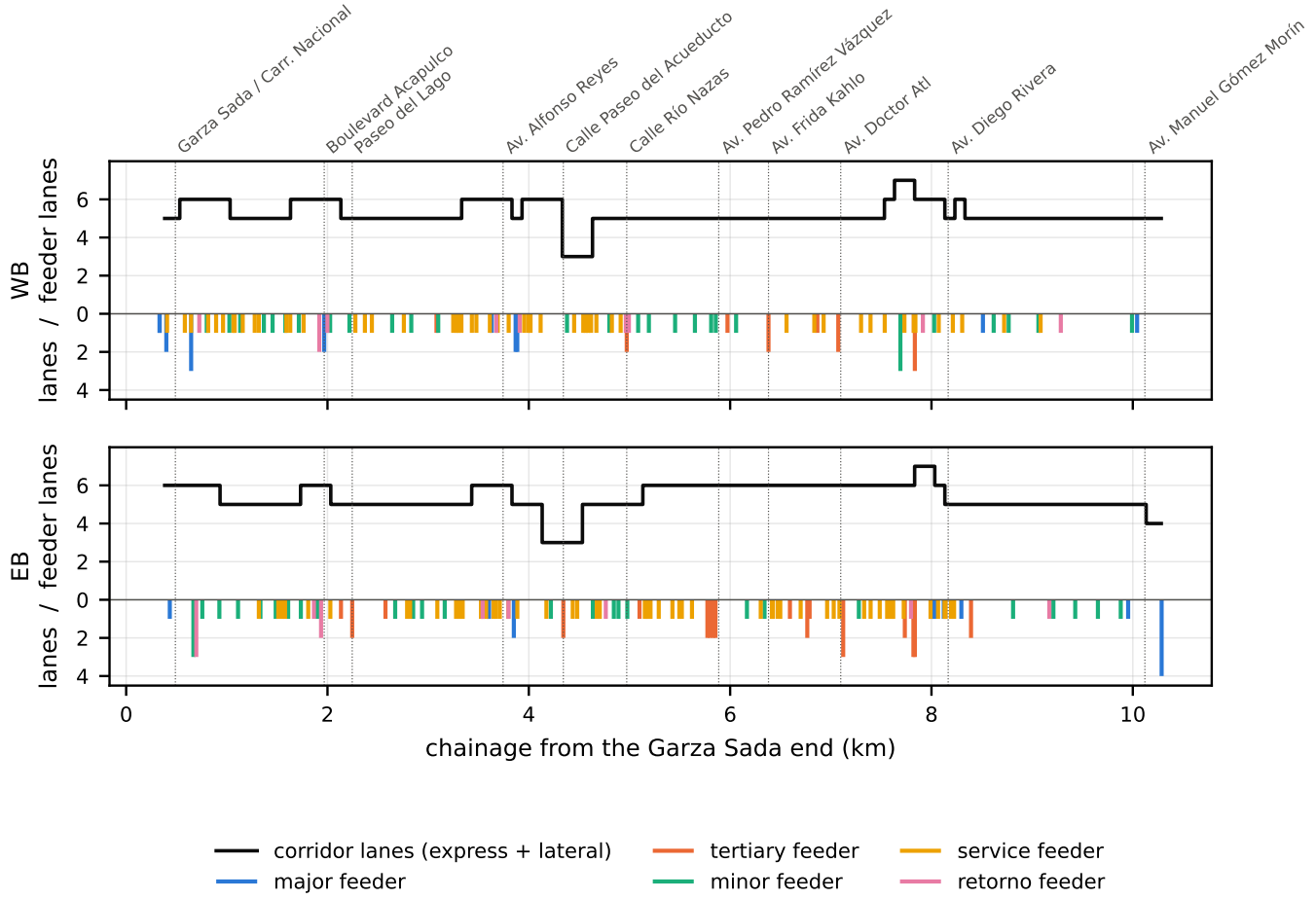


Figure 1: Inventory along the corridor, measured from the Garza Sada end (top: westbound, towards San Pedro; bottom: eastbound). Black: corridor lanes per direction (express plus frontage). Below the axis: every feeder approach, with stem length equal to its lanes, coloured by road class. Dotted lines mark the 11 junctions that split the corridor into segments. Note the drop to three lanes at 4.4–4.6 km.

street and its unnamed ramps, retornos excluded). The demand model had assumed 2,100 veh/h there, mostly from a two-lane slip ramp rated as a major feeder. The count replaces the assumption: the base run uses the middle of the range, the Monte Carlo draws inside it, and the continuous-turn lane bypasses the signal. The count records what entered, so it equals demand only if no queue remains on Alfonso Reyes after the green; it is an estimate, not a measurement.

4 Model

Cells and feeders. Each direction is a line of 100 m cells in travel order. The capacity of cell i's downstream boundary is

$$Q_i = s_0 \left(\ell_i^{\text{exp}} + \ell_i^{\text{lat}} g_i\right)\left(1 - f\, n_i^{\text{acc}}\right), \tag{1}$$

with saturation flow $s_0 = 1900$ veh/h/lane [18], express and frontage lanes ℓ, green ratio g_i at signalised cells (0.55 at base), and a friction loss f (1.5% at base) for each unsignalised access n_i^{acc} in the cell. Every approach is a point queue that discharges at most at its own capacity. That capacity is $s_0(1 - g - 0.1)$ per lane at a signal, 1600 veh/h per lane for a ramp, and 700 veh/h per lane for a give-way merge. Through traffic enters at the upstream end (3500 veh/h at base).

Demand. An approach delivers $q = \ell\, r_{\text{class}}\, \tau$ veh/h: its lanes, times a per-lane peak rate by class, times the fraction that turns onto the corridor ($\tau = 1$ when the street ends there, 0.30 when it crosses). Base rates are 800, 450, 100 and 25 veh/h/lane for major, tertiary, residential and service. We cannot measure these, so the Monte Carlo varies each one over a range roughly twice that wide (log-uniform). Departures attract vehicles by the same rule. Their split fractions β_i are scaled so that the vehicles leaving by side streets equal those that entered by them, which keeps corridor flow bounded.

Three levels. *Closed forms*: each lane group is a server. We use Erlang C, M/D/1, Kingman [11] and Webster [19], and validate them against a vehicle-level discrete-

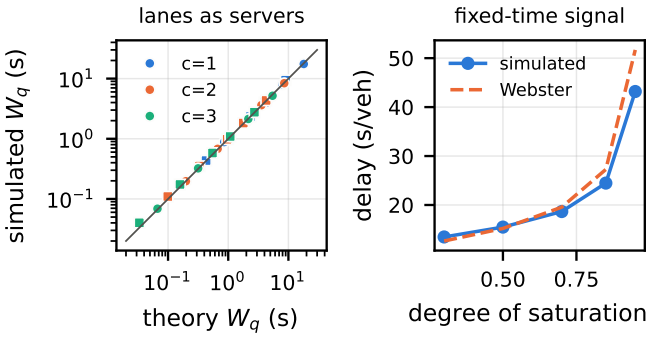


Figure 2: Validation. Left: simulated versus theoretical mean wait for 1–3 servers (lanes), Poisson arrivals, exponential (circles) and deterministic (squares) service, $\rho = 0.3$–0.9. Right: delay at a fixed-time signal (90 s cycle, $g/C = 0.5$) versus Webster.

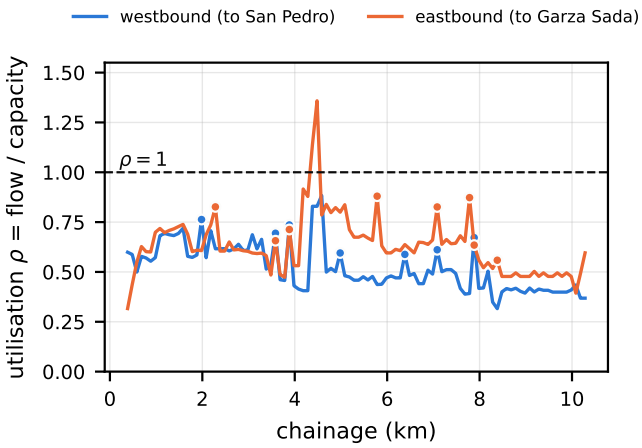


Figure 3: Steady-state utilisation along the corridor at base parameters; dots mark signalised cells. Both directions exceed $\rho = 1$ only at the five-to-three-lane merge near 4.5 km.

event simulator. *Tandem model*: the flow at cell k is $\lambda_k = \min(\lambda_{k-1}, Q_{k-1})(1-\beta_{k-1})+a_k$, where a_k is the flow from feeders that join at k, and utilisation is $\rho_k = \lambda_k/Q_k$. *CTM* [6, 7]: triangular fundamental diagram (60 km/h free flow, 20 km/h backward wave, 150 veh/km/lane jam density), $\Delta t = 6$ s. Feeders merge with Daganzo's priority rule (priority proportional to lanes). Diverges are partially FIFO: a fraction ϕ of exiting vehicles is held back when the continuing flow is blocked ($\phi = 1$ is strict FIFO, $\phi = 0$ lets exits leave freely; base 0.5). Discharge from a queued cell falls by a capacity drop of 8% [10, 5]. Demand follows a trapezoidal peak over a five-hour horizon: 45% of peak, rising for 30 minutes to a two-hour plateau, then falling. Total delay is the vehicle-hours spent in excess of free flow on the corridor plus all vehicle-hours waiting in feeder queues. An active bottleneck is a cell boundary with density above critical upstream and below it downstream [4].

Predictors. For each segment between junctions we compute, without simulation: the local lane ratio (feeder lanes entering the segment over its corridor lanes); the cumulative lane ratio (all feeder lanes up to the segment over its lanes, the localised "150 into 6" argument); feeder lanes alone (cumulative numerator); capacity alone ($1/\min Q$, geometry and signals without demand); and $\max \rho$. The ground truth is where the CTM places the principal active bottleneck.

5 Where the queue forms

Validation. The discrete-event simulator matches Erlang C, M/D/1 and Allen–Cunneen with a mean error of 3.2% over 30 settings (worst 20%, Allen–Cunneen at light load), and Webster's signal delay within 8% on average (Figure 2). Without capacity drop, the CTM's total delay behind a single lane drop equals the fluid point queue to within 0.00%, as Newell's equivalence requires [15].

The lane count overstates demand by an order of magnitude. If every feeder lane discharged at saturation, the 143 eastbound feeder lanes would deliver about 272,000 veh/h. Under the base rates they deliver 21,600 veh/h, or 151 veh/h per lane (8% of saturation; westbound 123 veh/h, 6%). Half the approaches are driveways and most of the rest are one-lane residential streets. Because exits balance the side inflow, peak corridor flow reaches only 7,800 veh/h eastbound and 6,700 veh/h westbound.

Utilisation picks one place. Figure 3 shows ρ along both directions. It stays below one everywhere except at the merge between the Alfonso Reyes overpass and the Arroyo Seco bridge, where it reaches 1.36 eastbound. Westbound it peaks at 0.88 at the same place. It would be 1.04 with the assumed demand at Alfonso Reyes instead of the field estimate. Signals raise ρ locally but do not push it over one at base parameters.

The peak. In the CTM (Figure 4) the westbound direction stays free-flowing at base demand. With the assumed demand at Alfonso Reyes it would queue for 2.1 km and lose 1,360 veh h, so one roadside estimate is enough to remove a predicted queue. The eastbound queue forms at 4.4 km, fills the 6.0 km upstream to Gómez Morín and costs 15,240 veh h (10.0 min per vehicle). A single-bottleneck fluid queue at the merge would predict 10,160 veh h. The difference comes from the queue reaching feeders and exits upstream. That is the most uncertain part of the model: with exits that leave freely ($\phi = 0$) eastbound delay is 4,170 veh h, and with strict FIFO it is 61,510. The location of the head does not move.

Sensitivity to demand. Scaling all demand (Figure 5), congestion appears at 0.8× base eastbound and

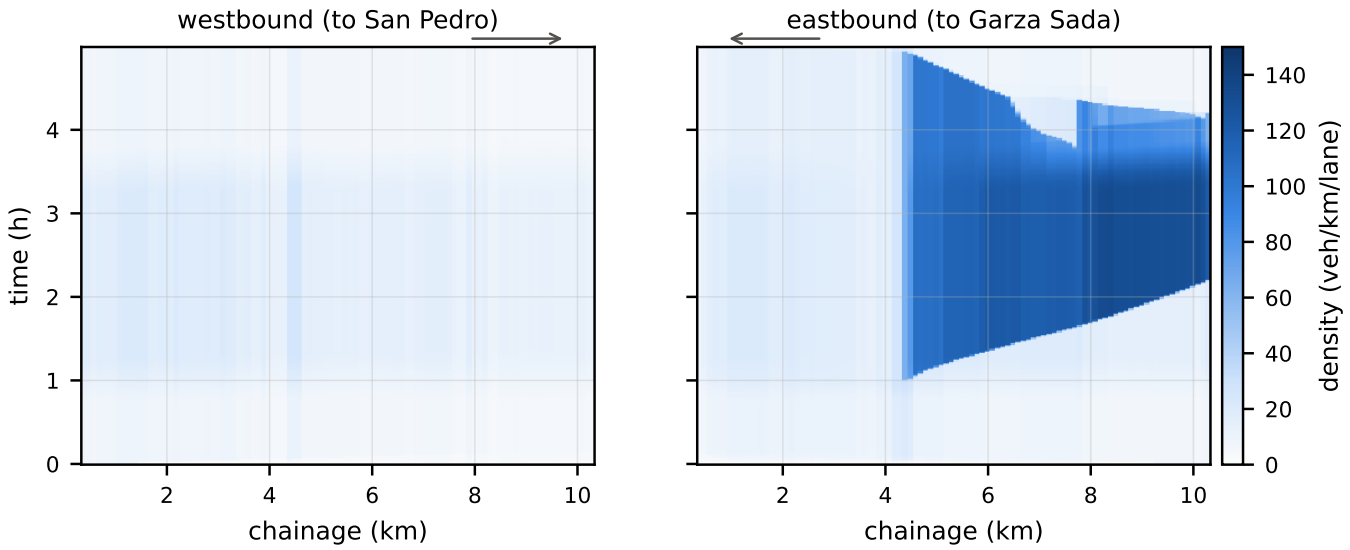


Figure 4: CTM density over time and space at base parameters, both drawn with chainage increasing to the right (arrows give the travel direction). Eastbound the queue head stays at the merge (4.4 km) and backs up across the whole upstream corridor to Gómez Morín; westbound, with the Alfonso Reyes field estimate, no queue forms.

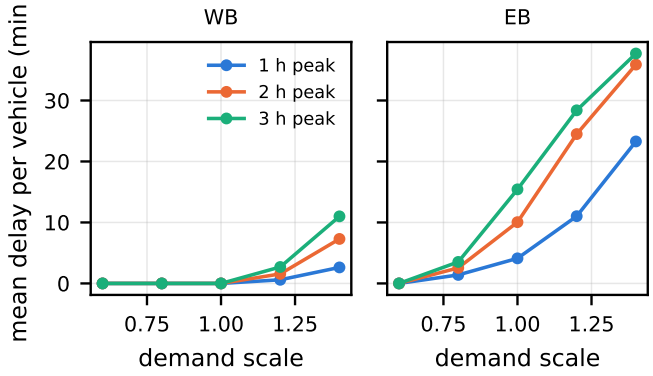


Figure 5: Mean delay per vehicle versus a uniform demand scale, for peaks of 1–3 hours.

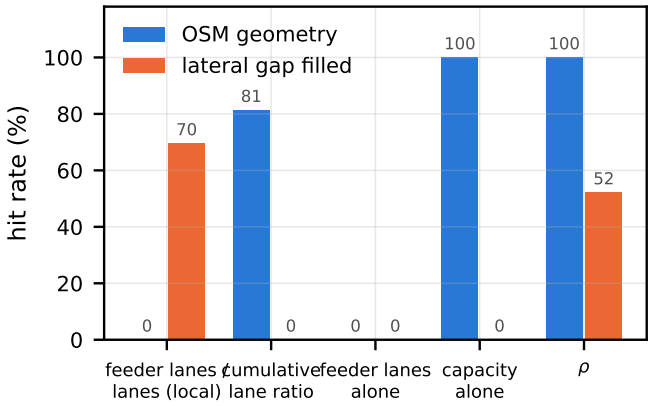


Figure 6: How often each predictor's top segment is the CTM's principal bottleneck, over 48 Monte Carlo draws × 2 directions (congested runs only), with the OSM geometry and with the frontage gap filled.

1.2× westbound, where the merge has about 10% spare capacity. At 1.4× with a two-hour peak, mean delay reaches 36 minutes per vehicle eastbound. Delay grows faster than linearly with peak length, as the fluid model predicts for a queue that grows throughout the peak.

The test. Over 48 Monte Carlo draws of every assumption (class rates, turn fractions, through demand, green ratio, friction, ϕ and capacity drop), 59 of 96 direction-runs congest (Figure 6). The local lane ratio picks the right segment in 0% of them, and feeder lanes alone in 0%: the segments with the most feeders are not where the queue starts. The cumulative lane ratio reaches 81%, but only through its denominator, which drops at the merge. Utilisation reaches 100%. Its maximum also classifies congestion: $\max \rho \geq 1$ agrees with the CTM in 100% of runs (AUC 1.00). The lane ratio cannot do this at all, since it does not change with demand. The hypothesis survives the falsification rule (falsified: no).

There is a caveat, and the Monte Carlo makes it plain: capacity alone also scores 100%. With this geometry, the location is set by the merge and demand only sets how bad it gets. To see what happens without it, we fill the frontage gap and rerun. Then only 23 of 96 runs congest, westbound congestion disappears at base parameters ($\max \rho = 0.76$; it does not congest at base parameters even with the gap), and eastbound the queue moves to 2.3 km. The location now depends on demand, and the ranking changes. Over the 23 congested runs (all eastbound, mostly one segment), the local lane ratio finds the bottleneck in 70% of them, utilisation in 52% and capacity alone in 0%. Without the merge, the queue starts where many accesses join within a short stretch. Friction and merge priority in the CTM pull the head there, while steady-state ρ, which treats each cell on its own, points to a neighbouring segment. So the lane-count

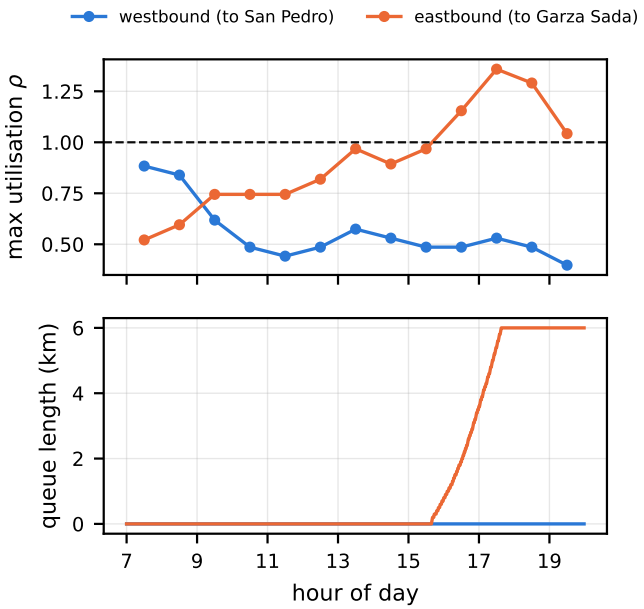


Figure 7: A working day, 7:00–20:00, with tidal demand (towards San Pedro in the morning, towards Garza Sada in the evening). Top: maximum utilisation along each direction, hour by hour. Bottom: queue length in the CTM.

argument is not wrong in general. On this corridor, the lane drop decides the location, and the lane count cannot see it.

A working day. The tests above use a generic two-hour peak. Commuting is tidal. We assume that at 7:00 traffic heads south to north, towards San Pedro and Valle Oriente, and that the evening reverses. The hourly demand multipliers are set in the configuration (Figure 7). The maximum capacity at the merge is about 5,360 veh/h westbound and 5,530 veh/h eastbound. With the Alfonso Reyes estimates, westbound utilisation peaks at 0.88 at 7:00 and no queue forms; across 24 Monte Carlo days, a westbound queue appears in 12%. Eastbound, ρ passes one in the afternoon. The queue forms at 15:44, reaches 6.0 km and is still there at 20:00, the end of the simulated day: 4.3 congested hours (median 3.9 over the Monte Carlo) and 17,060 veh h within the window. The observer reports little or no morning queue towards San Pedro, even though the traffic from Carretera Nacional heads that way. That is the one qualitative check we have against the field, and the model passes it. The eastbound afternoon queue has no field check yet.

6 What would shorten the queue

We apply four interventions around the principal bottleneck b in every congested Monte Carlo run (26 direction-runs) and compare total delay with the same run untreated (Figure 8).

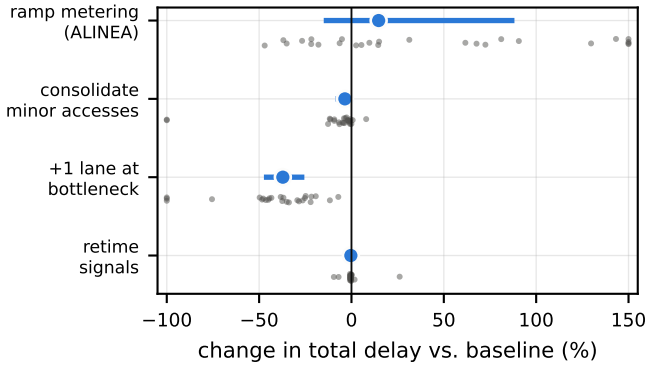


Figure 8: Change in total delay (corridor plus feeder queues) for four interventions at the CTM's principal bottleneck, over 26 congested Monte Carlo direction-runs. Blue: median and interquartile range; grey: individual runs (clipped at +150%).

One more lane over 600 m at the merge changes total delay by a median -37% and helps in 100% of runs. At base parameters it changes eastbound delay by -37%, after which the next-weakest section becomes the limit. In practice this means extending the frontage road through the gap, or adding an auxiliary lane on the express side. The model holds demand fixed, so it omits the extra traffic that new capacity tends to attract [8].

Ramp metering (ALINEA) on the four largest feeders within 1.5 km upstream moves delay rather than removing it: median change +15%, better in 35% of runs and worse in 65%. At base parameters eastbound, corridor delay falls from 8,920 to 4,340 veh h while feeder queues grow from 6,310 to 10,130. The net gain (-5%) comes from avoided capacity drop, and it depends on the drop being large. On an arterial whose "ramps" are residential streets, those waiting vehicles sit in neighbourhoods.

Consolidating minor accesses sends residential and service traffic of each segment to a collector with a single well-designed merge lane. This removes friction, but it also concentrates the side flow. The median change is -4% (better in 77%, worse in 4%). At base parameters it helps little eastbound (-9%).

Signal retiming (+0.10 green ratio for the corridor within 1.5 km upstream and 0.5 km downstream) barely matters: median -0.3%. Signals are not where the queue starts, and the green they gain is taken from cross streets.

7 Limitations and conclusion

Limitations. Apart from one roadside estimate at Alfonso Reyes, there are no traffic counts in this study. Demand is a parametric model, varied over wide ranges, and every absolute delay should be read in that light.

The conclusions we draw are the ones that survive the whole Monte Carlo: where the queue forms, and which predictors find it. The repository accepts field counts (`data/field_counts.csv`, vehicles per hour per approach and direction, or vehicles per minute on green and red); the Alfonso Reyes estimate shows how much one count can change: it removed the predicted westbound queue. Fifteen minutes of video at a few junctions, especially eastbound, would replace the largest remaining assumptions with data. OpenStreetMap is incomplete in two places that matter. Signals are under-mapped, so we infer them at major frontage junctions. The five-to-three-lane merge that dominates the results is a mapped feature that should be checked on site; if the frontage road actually continues there, the bottleneck moves, as the filled-gap experiment shows. The CTM aggregates express and frontage lanes into one cross-section, applies green ratios as average capacities rather than cycles, treats each approach as a vertical queue, and gives every vehicle the same route behaviour. How strongly the queue blocks exits (ϕ) changes delay several-fold and is the first thing a field study should measure. The model has no route choice, so traffic does not shift to Gómez Morín, Vasconcelos or Garza Sada when Lázaro Cárdenas fills up.

Conclusion. The lane count behind the popular explanation is right: about 133 street lanes, 264 with driveways, into 5 per direction. But it is the wrong quantity. Feeder lanes carry a few percent of their saturation flow, and their number does not locate the queue: the local lane ratio found the bottleneck in 0% of congested runs. Utilisation found it in 100% and predicted whether congestion occurs at all. Remove the lane drop, though, and the lane ratio becomes the better locator (70% against 52%, in a small sample): where many accesses join in a short stretch, their friction matters. On this corridor, at this map, the answer is geometric: a merge from five lanes to three between Alfonso Reyes and the Arroyo Seco bridge. The intervention that helps in every run is capacity at that merge, not fewer driveways or retimed signals. The rule generalises beyond Monterrey: to find where a corridor jams, compare demand with capacity at each merge, and look first for the place where lanes disappear.

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